

Track 4 · Shopping Copilot · Astrid CLI Engine · Zero Token Cost

ShopPilot

Offline-first multi-turn shopping copilot

TechnicalScore: 0.907

Hit@10: 97.5%

MRR: 0.858

MTTC: 2.88 turns

github.com/algorathem/techjam2026-shopping-copilot

Why Static Keyword Retrieval Breaks in Conversational Search

Single-turn search assumes fixed intent. Real shopping sessions suffer from four core failure modes:

01

Vague queries

"something nice for summer" floods the returned results

02

Semantic Collisions

"dress sandals" pulling dresses instead of footwear

03

Mind-change

Mid-session pivots leave stale context in retrieval

04

Turn Fatigue

miss after turn 10 → cart abandonment

Objective Function

Find the hidden parent_asin within a 50k frozen catalog under the official formula:

$$\text{TechnicalScore} = 0.50 \cdot \text{Hit@10} + 0.30 \cdot \text{MRR} + 0.20 \cdot \left(\frac{11 - \text{MTTC}}{10} \right)$$

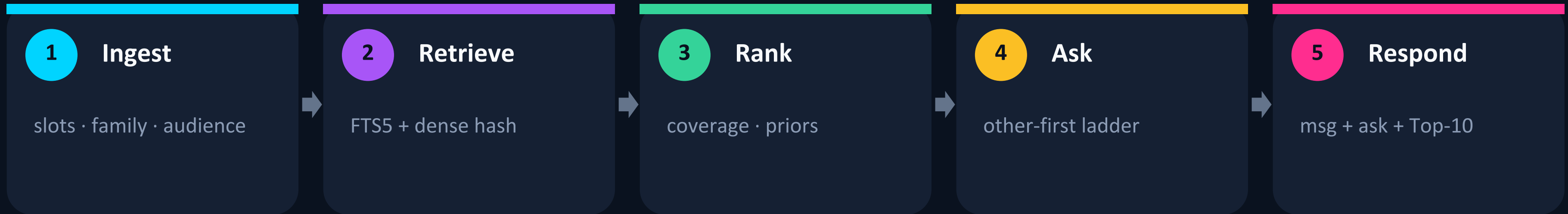
ShopPilot Engine: Headless, Fast, and Network-Free

Every respond() turn returns a natural message, one structured ask_attribute, and Top-10 ASINs in < 15ms.



Multi-turn loop — state is the product

Each turn: past SessionState + current text → Top-10 + one ask. Catalog immutable. LLM optional / off.



SessionState · mutable

soft | disclosed | override sources
family · audience · asked / filled
soft-only wipe on mind-change
next turn = same session_id

Catalog · immutable

FTS5 BM25 in-memory
dense hash 512-d default
MiniLM opt-in
50k CSJ freeze · no turn writes

Optional LLM · gated

slots NLU + top-20 rerank
fail-open timeout → rules
NOT on default score path
tokens on default path: 0

End-to-end system diagram

Native PowerPoint shapes — fully editable in Canva / Keynote / PPT · Tech 0.909 · Hit 0.975 · 0 tokens



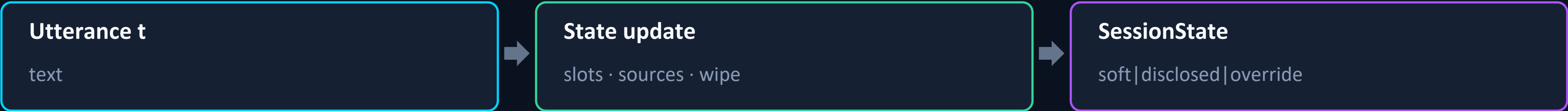
Classical CRS stack → ShopPilot modules

Editable shapes (not paper screenshots) · Qulac · Bayesian CRS · DST · corpus CQs · hybrid IR

A · CLASSICAL CRS PIPELINE



B · DIALOG STATE TRACKING



C · CLARIFYING QUESTIONS

Theory (ablated)

max E[IG] / min H(posterior)
Bayesian CRS · pool entropy
brand/store splits look high-IG
on THIS kit → Tech ~0.72 REJECT

ShopPilot ship path

other-first + static ladder
skip filled / dont_care
corpus facets in MESSAGE only
protocol-aligned CQ · Tech ~0.91

8.5× Lift Over Baseline

Evaluated deterministically via official evaluator.local_evaluator on the 50,000 Amazon CSJ dataset.

Metric	Weak BM25	ShopPilot (hash)	Delta / Lift
TechnicalScore	0.107	0.907	+0.800 (8.5×)
Hit Rate@10	12.5%	97.5%	+85.0 pp
MRR	0.068	0.858	+0.790 (12.6×)
MTTC (turns)	9.81	2.88	-6.93 (-70.6%)
LLM Token Cost	0	0	100% offline

METRIC TRADE-OFF · THE EARLY-TERMINATION PARADOX

Emitting Top-10 too early on Turn 1 risks a low-rank hit (e.g. Rank 8 MRR=0.125), freezing a poor score. Because MRR carries 1.5× the weight of MTTC in TechnicalScore, ShopPilot uses Precision Gating (emits only Top-1 on ambiguous early turns) to gather constraints and force final conversion toward Rank 1.

Live Execution: Astrid Multi-Turn Terminal

```
python3 cli_chat.py --dense hash · /new /state /quit
```

astrid — dense=hash

```
$ python3 cli_chat.py --dense hash
```

Astrid · quiet clarity for every aisle

You · shoes for my son

Astrid: Got it (footwear; boys). Any color...?

ask · other

You · Actually forget sneakers, dress shoes size 9

Astrid: Updated (footwear; size 9). Color?

soft wipe · disclosed kept · family=footwear

```
/state → size=9 asked={other}
```

Vague → Clarify

dress + exploring → ask other → "plus size" locks size (no re-ask).

Taxonomy Lock

"dress sandals" → family=footwear; suppresses garment dresses.

Preference Override

running shoes → dress shoes size 9; soft wipe; size persists.

Thank You

ONE-LINE REPRODUCTION

```
export SHOPPILOT_DENSE=hash; python3 -m evaluator.local_evaluator  
github.com/algorathem/techjam2026-shopping-copilot
```